

SE4041 – MOBILE APPLICATION DESIGN AND DEVELOPMENT
ASSIGNMENT 01 – PHASE 01 PROPOSAL



Community Services and Local Marketplace Mobile Application

Faculty of Computing
BSc (Hons) in Information Technology
Specializing Interactive Media
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Sandaruwan W J – IT22294548



1. INTRODUCTION & PROPOSED IDEA

This proposal presents සාමාජිකයා, a Kotlin-based Android mobile application designed under the assigned domain “Community Services and Local Marketplace.” The application connects people within a defined local area (neighborhood, apartment complex, university zone, or city ward) so they can request and offer community help, discover trusted local service providers, and buy, sell, or borrow goods safely.

1.1 WHY THIS IDEA?

In Sri Lanka and many similar markets, community help and local trade still rely heavily on WhatsApp groups, Facebook Marketplace, and informal word-of-mouth. These channels are noisy, hard to search, weak on trust, and not designed for both community service coordination and local commerce in one place. සාමාජිකයා fills that gap with a purpose-built hyperlocal platform focused on trust, proximity, and practical everyday usefulness.

1.2 APPLICATION VISION

Brand meaning: "Yalu" (from Sinhala yaluwa / friend) + "Way" (path). Together, යාලුway means a friendly local path to get help, offer services, and trade with people nearby. Tagline: "Help nearby. Trade nearby. Trust nearby." යාලුway aims to become the go-to mobile hub for neighborhood support and small-scale local exchange, combining mutual aid with a structured local marketplace.

2. PROBLEM STATEMENT

Current solutions create several practical problems for residents, students, and informal service workers:

- ✓ Community needs (blood requests, volunteer help, lost items, elder support, tool borrowing) are scattered across chat groups and social media posts that expire quickly.
- ✓ Local buying/selling is dominated by general classified platforms that are city-wide, low-trust, and not community-oriented.
- ✓ Skilled informal workers (tutors, electricians, cleaners, home cooks) struggle to get visible, verified local demand without paying for ads or relying on random referrals.
- ✓ Users have no single place to request help, offer services, and trade goods within a trusted radius.
- ✓ Safety and credibility are weak: limited verification, weak ratings for community help, and poor category structure.

යාලුway addresses these issues by providing one mobile application for community service posts, service-provider discovery, and a hyperlocal goods marketplace with profiles, categories, ratings, and local filters.





3. MARKET GAP ANALYSIS

Existing platforms partially cover this space, but none fully serve the hyperlocal community + marketplace combination for the Sri Lankan / South Asian neighborhood context:

- ✓ Facebook Marketplace / Groups: High usage, but cluttered, unsafe messaging, weak service booking, and poor discovery for urgent community needs.
- ✓ ikman.lk / classified sites: Strong for ads, weak for community mutual aid, volunteer matching, and neighborhood-level trust.
- ✓ TaskRabbit / Airtasker-style apps: Useful abroad for task hiring, but limited local presence, high formality, and not focused on community volunteering or goods exchange.
- ✓ Donation / volunteer apps: Often single-purpose (blood, charity) and do not support local trade or micro-services.

Market Gap: There is no widely adopted mobile application that unifies (1) community help requests, (2) verified local micro-services, and (3) neighborhood buy-sell-borrow trading with trust signals and location relevance. *සබ්වේ* targets exactly this underserved intersection.

3.1 OPPORTUNITY JUSTIFICATION

- ✓ Rising smartphone penetration and preference for mobile-first local discovery.
- ✓ Strong cultural habit of neighborhood mutual help that is currently digitally unstructured.
- ✓ Growth of home-based and informal services that need low-friction visibility.
- ✓ Student and apartment communities need safer, localized exchange than open public marketplaces.

4. TARGET AUDIENCE



4.1 PRIMARY USERS

- ✓ Residents of neighborhoods, housing schemes, and apartment complexes (age 18–45).
- ✓ University students living in hostels / boarding places who buy, sell, borrow, and need local help.
- ✓ Informal and semi-formal local service providers (tutors, repair persons, cleaners, bakers, caregivers).

4.2 SECONDARY USERS

- ✓ Community admins / apartment managers who moderate local zones.
- ✓ Volunteer groups and donation organizers coordinating short local campaigns.
- ✓ Elderly residents' family members arranging nearby assistance.

4.3 USER PERSONAS (SUMMARY)

- Nimali (24, student): Sells used textbooks, borrows a kettle, finds a nearby math tutor.
- Kasun (33, resident): Needs a plumber urgently and wants neighborhood-verified options.
- Amaya (29, home baker): Lists homemade cakes and accepts local pickup orders.
- Mr. Silva (68): Family posts a request for weekly grocery assistance from trusted neighbors.

5. FEASIBILITY STUDY

5.1 TECHNICAL FEASIBILITY

Fully feasible within the assignment scope using Kotlin and Android Studio. Core features (authentication, listings, requests, categories, search/filter, ratings, and local database CRUD) can be implemented with standard Android components and a local/offline-capable database such as Room (SQLite). UI can be prototyped in Figma and built with XML or Jetpack Compose layouts. No dependency on unavailable proprietary APIs is required for MVP.

5.2 OPERATIONAL FEASIBILITY

Users already understand posting, browsing, and messaging patterns from social apps. *සබ්ද්වය* simplifies this into clear modules: Community Help, Services, and Marketplace. Moderation can begin with basic report/block controls and admin flags for academic demonstration.

5.3 ECONOMIC FEASIBILITY

Development cost is low for a student MVP (no paid cloud requirement if local DB is used). Future monetization could include optional featured listings or small service boosting fees, but the academic version remains free and self-contained.



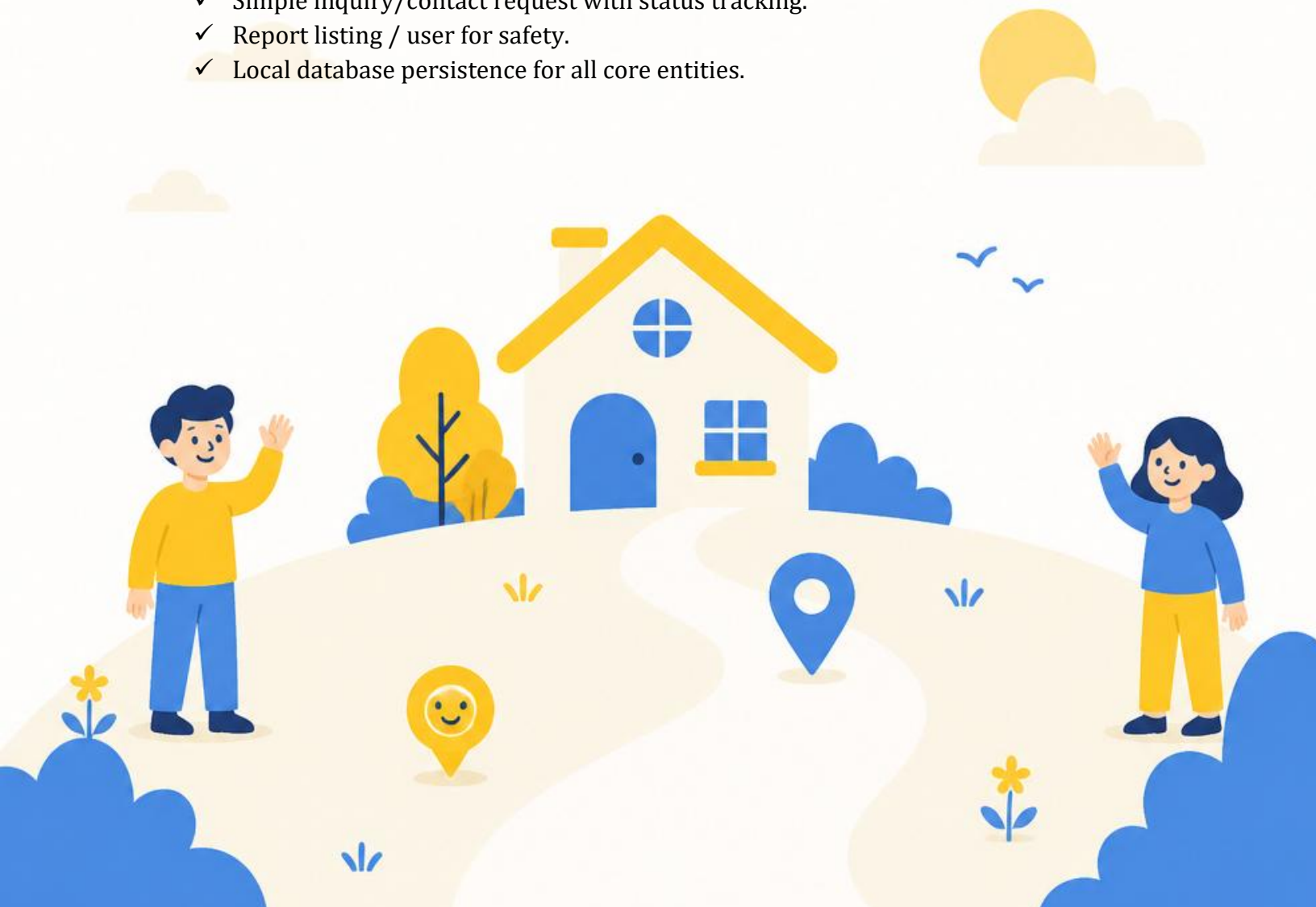
5.4 SCHEDULE FEASIBILITY

Aligned with SDLC stages required by the module: proposal approval → requirements → Figma UI → Kotlin development with GitHub commits → unit/integration testing → final documentation. Scope is intentionally MVP-focused to remain deliverable.

6. SCOPE OF THE APPLICATION

6.1 IN SCOPE (MVP)

- ✓ User registration / login and profile management.
- ✓ Create, view, update, and delete community help posts (request/offer).
- ✓ Local marketplace listings for goods (sell / donate / borrow).
- ✓ Service provider profiles and service listings by category.
- ✓ Search and filter by category, keyword, and locality/area tag.
- ✓ Favorites / saved items.
- ✓ Basic rating and review for completed help or services.
- ✓ Simple inquiry/contact request with status tracking.
- ✓ Report listing / user for safety.
- ✓ Local database persistence for all core entities.



6.2 OUT OF SCOPE (FUTURE)

- ✓ Real payment gateway / in-app checkout.
- ✓ Live GPS tracking of service providers.
- ✓ Advanced AI matching or recommendation engines.
- ✓ iOS version.
- ✓ Deployment to Play Store production release and long-term maintenance (excluded by assignment).
- ✓ Complex multi-admin enterprise moderation dashboards.

7. OBJECTIVES

- ✓ Provide a trusted hyperlocal digital space for community support and local exchange.
- ✓ Reduce dependence on unstructured social media groups for urgent neighborhood needs.
- ✓ Help local micro-service providers gain visible, category-based discovery.
- ✓ Enable safer buy/sell/borrow interactions within a defined community area.
- ✓ Demonstrate complete SDLC execution from planning to testing using Kotlin.

8. FEATURES AND FUNCTIONALITIES

8.1 CORE MODULES

- ✓ Auth & Profile: Sign up, login, edit profile, area/community selection, skills/tags.
- ✓ Community Help Board: Post requests (blood, volunteers, lost & found, elder help) and offers; mark as open/closed.
- ✓ Local Marketplace: List goods with images, price/free/borrow mode, condition, and area.
- ✓ Services Directory: Create service ads (tuition, repairs, cleaning, caregiving, homemade food) with availability.
- ✓ Discovery: Home feed combining recent help posts, services, and marketplace items near selected area.
- ✓ Trust Layer: Ratings, reviews, verification badge (manual/demo), report/block.
- ✓ My Activity: Manage my posts, saved items, and response status.

8.2 KEY DIFFERENTIATORS

- ✓ Three-in-one hyperlocal model: Help + Services + Marketplace.
- ✓ Community-area first design (not generic nationwide classifieds).
- ✓ Borrow/donate modes in addition to selling.
- ✓ Trust and safety controls are designed for neighborhood use.

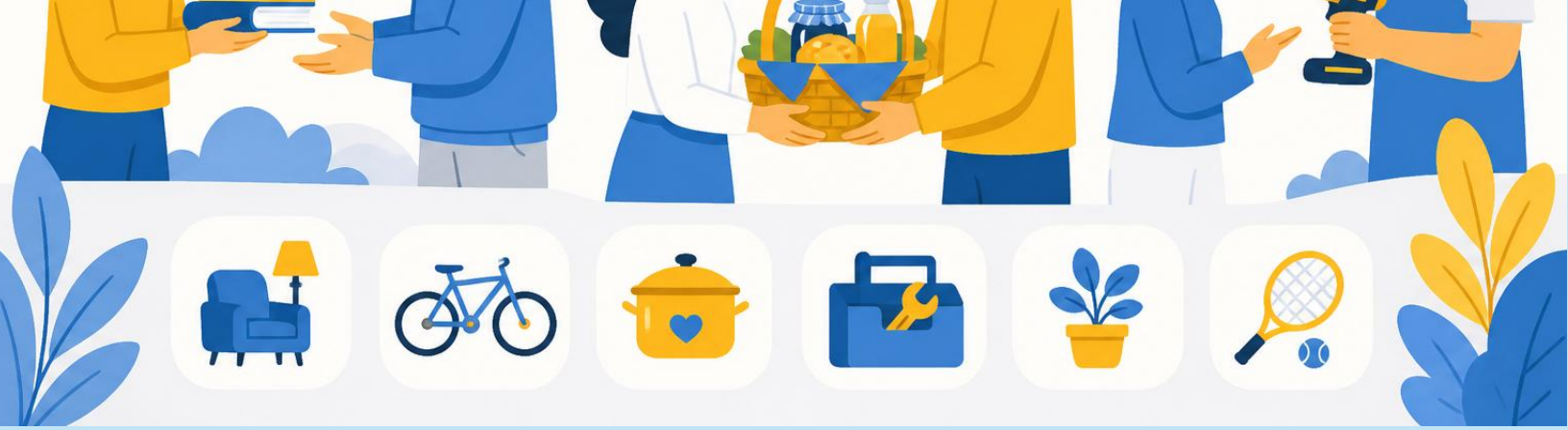


9. DEFINING REQUIREMENTS

9.1 FUNCTIONAL REQUIREMENTS

- FR01: The system shall allow users to register and authenticate securely.
- FR02: The system shall allow users to create and manage profiles including locality and contact preferences.
- FR03: The system shall allow users to create community help requests and offers with category and urgency level.
- FR04: The system shall allow users to publish marketplace listings (sell/donate/borrow) with details and images.
- FR05: The system shall allow service providers to publish and manage service listings.
- FR06: The system shall allow users to browse, search, and filter posts by category, keyword, and area.
- FR07: The system shall allow users to save/favorite listings.
- FR08: The system shall allow users to submit ratings and reviews after interactions.
- FR09: The system shall allow users to update listing status (available, reserved, completed, closed).
- FR10: The system shall persist all user and listing data in a local database.
- FR11: The system shall allow users to report inappropriate content.
- FR12: The system shall provide a dashboard/home screen summarizing nearby activity.





9.2 NON-FUNCTIONAL REQUIREMENTS

- NFR01 – Usability: New users should complete core tasks (post/browse) with minimal learning time; clear navigation.
- NFR02 – Performance: Common screens (feed, search results) should load smoothly on mid-range Android devices.
- NFR03 – Reliability: CRUD operations must consistently persist and retrieve data without loss.
- NFR04 – Security: Passwords stored using safe hashing; sensitive actions protected by login session.
- NFR05 – Maintainability: Modular Kotlin architecture (UI / ViewModel / Repository / DB layers).
- NFR06 – Compatibility: Support modern Android API levels used in Android Studio target devices/emulators.
- NFR07 – Accessibility: Readable typography, adequate contrast, and touch-friendly controls.
- NFR08 – Scalability (design-level): Data model should allow future migration to cloud backend.

14. CONCLUSION

သံဇာတံ is a practical, innovative, and academically achievable response to the assigned topic. It solves a clear local problem—fragmented community help and low-trust neighborhood trading—by combining community services and a local marketplace in one Kotlin Android application. The idea has a visible market gap, a well-defined MVP scope, and strong alignment with the SE4041 marking criteria for ideation, design, development, database use, and documentation.

